KUMARAGURU COLLEGE OF TECHNOLOGY

Department of Artificial Intelligence and Data Science

DATA SCIENCE AND VISUALIZATION (24ADI204)

WEEK 6 REPORT

Feature Engineering

Team 16: Youth Smoking and Drug Consumption Analysis

DATA-VIZ HACK-A-BIT 2026–2027

Team Member Name	Roll Number
Atchaya S	25BAD016

1. Week Contribution:

Reviewed statistical distributions before and after feature transformation and maintained the Week 6 notebook and related project files in GitHub.

2. Work Done:

During Week 6, my individual work focused on reviewed statistical distributions before and after feature transformation and maintained the Week 6 notebook and related project files in GitHub.

- Reviewed the numerical personality features selected for feature-engineering analysis.
- Studied scaling and transformation concepts and why they may be used to prepare numerical features for further analysis.
- Compared distribution characteristics before and after the transformation checks carried out in the project.
- Reviewed skewness values of the personality variables and the project decision on whether a transformation was required.
- Maintained the Week 6 feature-engineering notebook and documentation in the Team 16 GitHub repository.

3. Topics Covered:

- Feature engineering and numerical-feature preparation.
- Scaling and transformation concepts.
- Skewness and distribution comparison.
- Interpretation of before-and-after statistical results.
- GitHub maintenance of Week 6 notebook and documentation.

4. Outcome of My Contribution:

I gained a better understanding of how feature preparation can affect later analysis and why transformations should be selected based on the observed data. The Week 6 work was recorded in GitHub.

5. GitHub Repository:

Repository: https://github.com/atchayas-ai/DSV_Team_16_Youth_Smoking_Drug_Abuse

I helped manage and maintain the shared repository by keeping the relevant weekly documentation, notebooks, datasets and project materials organised. The exact files updated during each week are described in the Work Done section.

6. Reference:

Project repository: GitHub – Team 16, Youth Smoking and Drug Consumption Analysis.